



University of Vavuniya

Second Examination in Information Technology – 2022

First Semester – January/February 2024

IT 2133 – Advanced Web Programming (Practical)

Answer all Questions

Time Allowed : Two hours

Instructions:

- Source files, MySQL dump file for Database and layout *Figures* of the expected design (given in Reference Images folder) are stored in the Resources folder.
 - Conceptual design of web page navigation is given in the *Figure 10* and stored in Reference Images Folder.
 - Create a folder on your Desktop, name it with your index number and save all files into that folder. Use XAMPP to run the php applications and PhpMyAdmin to run the database required.
1. The University of Vavuniya's administration has decided to create an e-library specifically for their students. You are requested to design a basic online library management system using PHP, HTML, and CSS, considering the provided requirements.

(a) User types of this web application:

- i. Administrator
- ii. Student

[This question is continued on the next page]

(b) Develop a *Configuration* page that encompasses variables for database connection details (server name, username, password, and database name). This page serves the purpose of establishing a connection between the application and the database. Integrate this configuration file into various PHP files to facilitate the establishment of a database connection.

(c) Create an *Index* page as shown in *Figure 1* which has two links for Login and Register pages.

(d) Create a *Registration* page as shown in *Figure 2* which authorizes the student to create an account.

i. If the registration is successful, the student should be redirected to the Login page. (Refer *Figure 3*)

ii. Use form validation techniques on the registration form to implement the requirements given below (either client-side or server-side):

- Every field must be completed; none can be left blank.
- The name and username fields permit only letters and spaces; other characters are not accepted.
- The email field should be a valid format.
- The password must consist of 8 to 20 characters and include at least one uppercase letter along with one special character.

iii. If any of the above given (ii) constraints are violated, you should display the error message for the following cases:

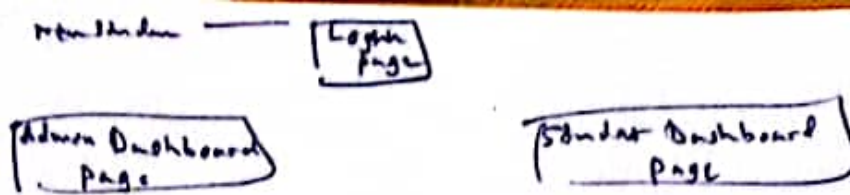
- A. Submit the form with an empty field.
- B. Submit the form with numbers or/and special characters in the name and username field.
- C. Submit the form with an invalid email format.
- D. Submit the form without fulfilling the specified password validation criteria outlined in (ii).

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Admin
Login
Student
Login
Link
Admin

1st Comp 1st (1st name) 1st
Refer 2nd

Insert data



(e) Create a login page as shown in Figure 3.

- i. If the login is successful, the Student should be redirected to the student dashboard page (Refer Figure 4) and the Administrator should be redirected to the admin dashboard page. (Refer Figure 5)
- ii. If the login is not successful, an Error message should be displayed.

(f) Create the web pages to achieve the tasks that the student and administrator follow.

A Student :

- i. The student can view the list of available books as shown in Figure 4 when the student clicks on the link "Click to Read" they can read the book.
- ii. When the student clicks on the link "Request Book", it will be redirected to the "Request Book" webpage where the student can request the needed books. (Refer figure 5)
- iii. When the student clicks on the link "Change Password", it will be redirected to the Change Password web page (Figure 6) where the student can change their current password.

When a student attempts to change their password, the system should follow the validation process.

Validation: The password should consist of 8 to 20 characters and include at least one uppercase letter along with one special character.

An administrator:

- i. An administrator can view all the uploaded book details on the administrator dashboard page. (Refer Figure 7)
- ii. When an administrator clicks on the link "View Requests" he/she can view all the requests from the students. (Refer Figure 8)
- iii. When an administrator clicks on the link "Upload Books" he/she can upload the student-requested new books to the e-Library system. (Refer Figure 9)

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When an administrator attempts to upload a book, the system should execute the following validations:

- The system should verify whether the book has already been uploaded into the database or not.
 - Only PDF books should be allowed for upload by an administrator.
- (g) Create a "logout" link for all the web pages except the registration page and login page.

Note:

- You can use the administrator login credentials:
Username - admin
Password - Admin@123
- Implement session handling concepts to prevent students from accessing pages that are exclusively accessible by the administrator.
- The database file contains essential information that can be utilized to implement.

[100%]